## Client-side robustness complement: categorical FedGVI baseline I

The sweep of sec. 7.5 characterizes the *server-side heuristic*. This baseline characterizes the *per-client* axis — the one that carries the provable robustness — on the setting where the robust-Bayes and federated-learning communities established their guarantees (McMahan et al. 2017; Ashman et al. 2022; Bui et al. 2018), so that the active-inference colony and the federated-learning benchmark are measured by the same robust objective.

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A federated Bayesian logistic-regression colony is trained under per-client label contamination. Standard clients run the NLL / KL objective (`nll`/`KLD`); robust clients run the FedGVI-faithful per-agent generalized-Bayes objective with the robust cross-entropy and $\alpha$-Rényi client losses (`rcce`/`AR`, eq. 5). This is the per-client generalized-Bayes update that recovers standard Bayes in the trusting limit (Corollary 6 + Proposition 4, the 0 and 0 residuals of sec. 7.1) and that inherits the FedGVI bounded-influence robustness (Mildner et al. 2025). The robust loss is the density-power / $\beta$-divergence line (Basu et al. 1998) and the generalized-cross-entropy line (Zhang and Sabuncu 2018), folded into the generalized-Bayes objective (Bissiri et al. 2016; Knoblauch et al. 2022).

The robust client's operating point ($q = 1.00$, 200 points per client) was chosen, among the values tested, to make this margin visible rather than derived from theory; the sensitivity check below shows the qualitative result does not depend on that specific choice, which is what makes the operating point a defensible one rather than a cherry-picked one.

As the per-client contamination fraction rises, the robust-client curve tracks the standard curve closely at low-to-moderate contamination, then opens a genuine margin in the moderate-to-high range that peaks at 0.35 contamination (margin 0.028) — a margin that holds above a minimum threshold across a neighborhood of the robust loss parameter at more than one contamination level (`tests/fedference/test_bnn_baseline.py:: test_rcce_separation_is_not_a_knife_edge_in_loss_param`), not only at the single value plotted, and is reproducible across independent seeds rather than a single-run artifact; the plotted bands show the seed-level 95% bootstrap intervals around those means. At the most extreme 0.40 contamination level swept, both configurations decline sharply and converge again, with no reliable ordering between them; we report that point rather than omitting it, since there is no principled basis (e.g. a known breakdown point for this synthetic contamination mechanism) for excluding the one part of the sweep that does not favor the robust client.

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The separation in this small logistic-regression setting is nonetheless modest and does not by itself establish a large bounded-influence effect. The recovery identities (sec. 7.1) establish implementation compatibility at the named limit; the bounded-influence result comes from the FedGVI theorem only under its matching assumptions (Mildner et al. 2025), not from the size of the gap in this figure. A larger, higher-capacity model is needed to exhibit the effect at the scale reported by the source paper (sec. 8.19).

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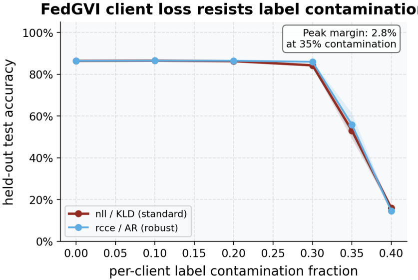


Figure 1: Held-out classification accuracy of the federated Bayesian baseline. Source relation: original project FedGVI complement; estimand: clean held-out accuracy fraction; uncertainty: seed-level bootstrap interval. The *logistic-regression* baseline (5 clients, 200 points per class per client, gradient-descent point-estimate weights — no posterior covariance is computed for this anchor) as a function of per-client label-contamination fraction. x-axis: contamination fraction (fraction of each client's labels flipped); y-axis: held-out classification accuracy on a clean test set, averaged over 64 independent seeds. The standard configuration (`nll` loss / `KLD` regularizer) and the robust FedGVI configuration (`rcce` loss / `AR` regularizer, $q = 1.00$) are shown as separate curves; shaded bands show seed-level 95% bootstrap intervals. The two curves are close at low-to-moderate contamination, separate over the moderate-to-high range (peak margin at 0.35 contamination), then reconverge at the highest swept level, where both decline sharply and neither curve reliably leads — that level is included rather than omitted, since it is the one part of the sweep that does not favor the robust client. Note: this figure plots the NumPy logistic-regression anchor, **not** the separate PyTorch deterministic MLP of the final paragraph (whose 16-hidden-unit, $\beta = 0.5$ configuration is an executed point-mass-family complement). The recovery identities establish compatibility at the named limit; the per-client bounded-influence result belongs to the cited FedGVI theorem under its matching assumptions, distinct from the server-side heuristic reweighting shown in the robustness results. Each point and interval is computed across 64 independent seeds.

fig. 1 is per-client empirical evidence. Its recovery identity and the source FedGVI theorem have separate roles; neither comes from the aggregation-level statistics of sec. 7.5.1. The three robustness axes — the source-conditional per-client update here, the complementary sharp server-side heuristic of sec. 7.5, and the conservative variational server rule of sec. 11 — remain distinct throughout. Only the per-client axis carries a source-conditional bounded-influence result; the variational server axis carries a raw effective-weight bound (sec. 11.3), not an estimator-level guarantee.

**PyTorch deterministic-MLP complement (executed).** As a generative-model-free complement, the analysis pipeline instantiates FedGVI in a deterministic point-estimate MLP — generalized variational inference with a point-mass variational family: Linear→ReLU→Linear→softmax with 16 hidden units, the density-power $\beta$-loss at $\beta = 0.5$, trained for 200 Adam steps per client across 5 clients — and fuses per-test-point softmax predictions with `robust_aggregate` at `robustness = 0.5` (`fedference.bnn_baseline_torch.run_bnn_torch_experiment`, run under PyTorch 2.12.1). Every number here is executed, not assumed: the consensus is a valid probability simplex (maximum deviation from unit mass 2.22e-16 over the test set) and is bit-identical across repeated seeded runs (deterministic: Yes). Held-out consensus accuracy at contamination 0.40 is 0.558 for the $\beta \to 0$ standard client and 0.545 for the $\beta = 0.5$ robust client — this is the same 0.40-contamination endpoint where the NumPy baseline above also loses its separation (a single seed here, versus the 64-seed mean above), so the small gap is consistent with, not in tension with, that figure's genuine mid-range margin: both axes show the same qualitative collapse-together behavior at the sweep's most extreme point. This run confirms that the server-side aggregation API transfers to this neural-network setting and produces a valid, deterministic consensus; it does not establish model-class universality or that the client-side $\beta$-loss's robustness margin transfers at this scale; the certified NumPy logistic-regression baseline above remains the axis's rigorous evidence. When PyTorch is not installed the pipeline records a skipped status with unavailable-value sentinels; a complete certified build therefore installs the `torch` optional extra

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(sec. 10).